TideTail: Evaluating Extreme-Aware Diffusion Augmentation for Coastal Water-Level Forecasting

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November 1, 2026
Preprint — evidence-linked, not all hypotheses verified

Abstract

Coastal water-level forecasting faces a fundamental data scarcity problem: the most operationally critical extreme storm events are rare in historical records, yet predicting them correctly is the primary goal of coastal risk management. We investigate whether conditional diffusion models can generate physically plausible extreme coastal water-level residual samples that improve downstream forecasting on independent real storm events, under matched forecaster training budgets. This paper reports the pre-registration of this investigation, the frozen data protocol, the engineering implementation, and the negative result of our first controlled diffusion pilot. The 27,169-parameter conditional DDPM trained for three epochs on real residual data produces samples whose standard deviation is approximately **22.95 times** that of the training target, with magnitudes exceeding ±16 metres. We trace this failure through independent CPU reproduction, terminal-step noise sensitivity analysis ($\alpha_{\text{bar}} \approx 5.93 \times 10^{-7}$), and end-to-end audit of the forward and reverse stochastic equations. The diagnosis confirms that while the model architecture and diffusion mathematics are formally correct, terminal-step amplification combined with imperfect epsilon prediction at the training budget causes catastrophic scale explosion. We present this finding, our repaired data-splitting infrastructure, and our frozen research protocol as a methodological contribution to the study of diffusion models for extreme-event time-series generation.

1 Introduction

Coastal flooding causes approximately $1 trillion in global damages annually and displaces hundreds of thousands of people [Yang et al., 2023]. Reliable forecasting of extreme water levels during storm events is therefore an urgent public safety problem. The core challenge is *data imbalance*: the most operationally critical storm surges are also the rarest. Training standard neural forecasters on historically recorded water levels allocates most model capacity to predicting the frequent benign conditions rather than the infrequent but catastrophic extremes.

Data augmentation has emerged as a promising approach to rebalance training distributions. For time-series forecasting, augmentation traditionally takes the form of oversampling, bootstrapping, noise perturbation, or interpolation. These methods are well-established for image data but have received less scrutiny for coastal water-level prediction, where physical laws (tidal cycles, wave dynamics, storm surge propagation) impose strong constraints on what constitutes a plausible augmented sample.

Conditional diffusion models [Ho et al., 2020] offer a generative framework that could produce physically realistic extreme events by learning the conditional distribution of water-level residuals given historical gauge observations. The hypothesis driving this work is:

> **H1**: Tail-aware conditional diffusion augmentation, applied under a matched forecaster training budget, improves prediction accuracy on *independent held-out real storm events* relative to real-only training, standard oversampling, perturbation augmentation, and standard conditional diffusion.

A secondary hypothesis (H2) investigates whether adding physical consistency or multi-gauge spatial coherence further improves synthesis usefulness. Both are *unverified hypotheses* at this writing. This paper reports the frozen research protocol, the engineering implementation, and the first controlled empirical evidence — which is, at this stage, a *negative result*: our pilot DDPM produces samples of physically implausible magnitude. We argue that this finding, its diagnosis, and the methodological infrastructure built to obtain it are themselves a contribution to the study of diffusion models for extreme-event time-series generation.

1.1 Contributions

1. We present a fully documented, leakage-safe research protocol for coastal water-level forecasting with diffusion augmentation, including event-grouped chronological splits, full-support embargo, train-only thresholds, and a six-region data manifest frozen at license-admission boundaries.
2. We implement and audit a conditional denoising diffusion probabilistic model (DDPM) for residual water-level generation, with 27,169 parameters, a cosine noise schedule, and 64 reverse diffusion steps.
3. We report a *reproduced negative result*: independently verified on both CUDA and CPU, the model's 24-hour generated residuals span -16.91 to $+7.94$ metres (SD 4.47m), approximately $22.95\times$ the training target standard deviation of 0.215m. We diagnose terminal-step amplification and imperfect epsilon prediction as contributing factors.
4. We present engineering validation guards (nonfinite rejection, train-reference scale screen) that prevent absurdly bad samples from being misclassified as successful generation. These guards are validation infrastructure, not model repairs.

1.2 Paper structure

Section 2 surveys related work. Section 3 details our frozen research protocol and data admission. Section 4 describes the conditional DDPM architecture and training setup. Section 5 describes our experimental plan and implementation. Section 6 reports our empirical findings, including the negative result of the pilot. Section 7 discusses implications, limitations, and ongoing work. Section 8 concludes.

2 Related Work

2.1 Coastal water-level forecasting

Traditional coastal water-level forecasting combines deterministic tidal constituents with empirical or dynamical storm-surge models [Yang et al., 2023]. In recent years, neural approaches have

improved forecast quality on held-out gauges. Sergeev et al. [2024] demonstrated that information from multiple gauges along a coastline substantially improves forecast accuracy for downstream stations. Tong et al. [2024] introduced dual-timescale residual memory that separately models slow tidal evolution and fast storm-driven residuals, with particular attention to extreme events. These works establish that (a) the de-tided residual signal carries the most operationally relevant information during storms, and (b) extreme events demand model architectures that differ from those optimized for benign conditions.

2.2 Data augmentation for time-series forecasting

Data augmentation for time series is reviewed by Hoppe et al. [2024] and related surveys. Standard approaches include: (1) *oversampling* rare events or weighting them in the loss function, (2) *perturbation* with additive or multiplicative noise, (3) *interpolation* between adjacent real windows, and (4) *generative models* including GANs, VAEs, and diffusion. A key distinction for coastal water levels is whether augmentation preserves the *physical plausibility* of the residual signal: unrealistic samples may actually degrade downstream forecasting by corrupting the training distribution.

2.3 Conditional diffusion models for time series

Conditional score-based diffusion models have been applied to time-series imputation [Tashiro et al., 2021, Ma et al., 2023] and generation. The canonical formulation [Ho et al., 2020] defines a forward process $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$ and trains a network $\epsilon_\theta(x_t, t, c)$ to predict the injected noise, where c denotes conditioning information. During generation, the reverse process iteratively denoises from T back to 0:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t, c)\right) + \sigma_t z, \tag{1}$$

where $\sigma_t^2 = \beta_t(1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t)$ is the posterior variance and $z \sim \mathcal{N}(0, I)$.

2.4 This work's niche

To our knowledge, this is the first systematic application of conditional diffusion augmentation to coastal water-level forecasting with (a) event-grouped leakage-safe splits, (b) train-only threshold estimation, (c) matched-budget comparison across augmentation methods, and (d) pre-registration of hypotheses before running experiments.

3 Research Protocol

3.1 Data source and admission

The authorized owner copy of the dataset is at `/data/yingda/tidetail-data/prime-formal-2015-2024`, containing 350 files totaling 451 MB. Six coastal regions are included, with 87,672 hourly UTC rows each spanning 2015-01-01 to 2024-12-31. Region admission status is recorded in `region_availability.json`:

The Hong Kong region is excluded from all formal experiments until the data owner confirms license, vertical datum, and the HK event `pearl_river_hong_kong_event_01149`, which spans the validation/test boundary at index 78904. The migration manifest (`MIGRATION_MANIFEST.json`) exists one level above the release directory; file SHA256 and size checks confirm all inspected files match both release and migration inventory entries.

Table 1: Region formal training authorization status

Region	Training authorized
New York Bight	Yes
Thames / North Sea	Yes
Hong Kong Pearl River	**No** (license/datum not signed)
Southeast Australia	Yes
US Gulf Coast	Yes
Charleston Atlantic	Yes

3.2 Target definition

The release manifest `database_manifest.json` defines:

- **Total water level**: observed water level above local chart datum.
- **Tide**: deterministic FES2022 tidal constituent forecast (no future uncertainty in this archival release).
- **Residual**: water_level_m = observed_total − FES2022_tide.

The pilot diffusion model targets `water_level_m` (residual), in metres, with a fixed 72-hour history → 24-hour forecast window for engineering feasibility. The formal release contract specifies 96-hour history → 72-hour forecast; all comparisons between methods should use one consistent contract for scientific validity.

3.3 Leakage-safe splitting

Our protocol enforces three hard gates before any window creation or normalization:

1. **Event-group disjointness**: the same independent storm event may not appear in two partitions. Events are represented as `EventGroup(event_id, start_idx, end_idx)` intervals. The implementation in `src/tidetail/splits.py` validates non-overlapping intervals and unique IDs before splitting.
2. **Chronological ordering**: train row block must end before validation row block, which must end before test row block. The split is 8:1:1 by chronological event-group order, subject to `min_events_per_split`.
3. **Full-support embargo**: for forecast origin time o, lookback L, and horizon H, the full temporal support of one forecast window is $[o-L, o+H)$. Two partitions must not have overlapping full-support intervals. The repaired `embargo_filter` compares every index in an event group to every protected index, replacing the earlier midpoint-only comparison that incorrectly accepted dangerous events (see Appendix A). Additionally, `embargo_full_support_filter` enforces $o \geq$ split_block_start $+ L$ and $o \leq$ split_block_end $- H$ for each forecast origin.

All scaler fitting, quantile threshold estimation, and interpolation fitting are performed on *train rows only* and then applied to all partitions without re-fitting. Threshold-run events crossing the train boundary are removed in full before window creation, not partial-interval trimmed.

3.4 Formal release contract

The database manifest specifies:

- **Input**: 96 hours of historical observations.
- **Horizon**: 72 hours of forecast.
- **Split**: chronological 8:1:1 with strict window containment.
- **Tide**: existing local FES2022 deterministic archive.
- **Forcing**: ERA5 hindcast only. No operational 72-hour forecast claim without archived issue/valid/lead NWP forecast vintages.

The ERA5 forcing data available in this release is historical hindcast data, not forecast data. Using future observed ERA5 values as conditioning information in a forecast experiment would constitute leakage unless explicitly labelled as an oracle experiment. Our gauge-only diffusion model does not consume atmospheric forcing; this limitation is discussed in Section 7.

4 Method

4.1 Conditional DDPM architecture

We implement a 1D temporal denoising diffusion probabilistic model with the following components:

1. **Temporal embedding**: sinusoidal frequencies $f_j = 10000^{-2j/D}$ embedded into a width-`width` MLP, where D is half the width.
2. **History conditioning**: a separate MLP consumes all 96 past observations (72 in the pilot), producing a `width`-dimensional conditioning vector that is *added* to the temporal embedding before entering the denoiser.
3. **Dilated residual blocks**: four blocks with dilation $\{1, 2, 4, 8\}$, each containing GroupNorm (groups=4), SiLU activation, 3-conv padding to preserve length, and a linear condition injection.
4. **Output head**: GroupNorm + SiLU + 3-conv producing a single-channel epsilon prediction at each of the 24 (pilot) or 72 (main-task) forecast horizons.

Total parameter count: 27,169 (pilot 72→24). For the main task 96→72, the horizon changes to 72 and the conditioning input to 96 values; parameter count remains in the 30–40K range, matched to the GRU and TCN budgets below.

4.2 Noise schedule and forward process

We implement cosine cumulative noise schedule [Ho et al., 2020] with $T = 64$ diffusion steps in the pilot:

$$\beta_t = 1 - \frac{\cos((t + 0.008)/1.008 \cdot \pi/2)^2}{\cos((t - 1 + 0.008)/1.008 \cdot \pi/2)^2}, \qquad \alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i. \tag{2}$$

At $t = T = 64$, the historical schedule reaches $\beta_{64} \approx 0.999$ and $\bar{\alpha}_{64} \approx 5.93 \times 10^{-7}$, producing $\approx 1299\times$ clean amplification at the final reverse step. An oracle comparison (Section 6) shows that subsampling a $T' = 1000$ cosine schedule to 64 steps reaches $\bar{\alpha}_{64} \approx 5.89 \times 10^{-4}$ (41× amplification), which eliminates the catastrophic scale explosion.

4.3 Training objective

Training targets *unit Gaussian epsilon* at each timestep. The loss is mean squared error between the network's epsilon prediction and the actual injected Gaussian noise:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon\sim\mathcal{N}(0,I)}\left[\left\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t, \text{history})\right\|^2\right]. \tag{3}$$

Training hyperparameters for the pilot run (Issue #6): AdamW with lr = 0.001, batch size 128, three epochs, BF16 precision on RTX3090 GPU 4, seed 20261025.

4.4 Forecasting baselines

All baselines consume 96 hours of hourly history and produce 72 hours of forecast, using the same train-only normalization as the diffusion model. Missing-value masks are concatenated as an extra input channel.

Rule-based baselines. *Persistence* carries forward the last observed water level for all 72 hours. *Seasonal baseline* repeats the most recent semi-diurnal tidal cycle (144 hours ≈ 6 days) to capture the dominant tidal periodicity.

GRU encoder-decoder. A 2-layer GRU encoder maps the 96-hour history into a hidden state; a GRU decoder then autoregressively produces 72 forecast steps, each step feeding its own output back as input. Hidden size is matched to DDPM parameter budget (≈ 5–10K). The GRU operates on normalized values with explicit gradient tracking for training and a no-grad `predict_eval` path for inference.

Temporal Convolutional Network. A stack of dilated causal 1D convolutions with Chomp1d padding ensures causality. Dilation grows exponentially ($\{1, 2, 4, 8\}$) to achieve a receptive field covering the full 96-hour history. The final hidden-state channel at the latest timestep is projected to all 72 horizons *at once* (non-autoregressive). Channel count is tuned to match DDPM budget (≈ 15–20K).

4.5 Training-set augmentation

All augmentations modify the training set only. Validation and test data are never augmented. Three augmentation strategies are implemented:

1. **A1 — Balanced extreme-event oversampling.** Windows where any target cell exceeds the train-only q95 threshold receive a higher sampling weight (≈ 3×) in each optimizer step. This addresses the rare-event training imbalance without fabricating new observations. Implemented via `torch.utils.data.WeightedRandomSampler`.

2. **A2 — Residual noise perturbation.** Training history (not target) is perturbed with additive Gaussian noise whose standard deviation is 30% of the train-only residual standard deviation. This models instrument uncertainty and sub-hourly unresolved variability. Targets are *never* modified.

3. **A2 — Tidal phase shifting.** Both history and target windows are circular-shifted by a random fraction of the semi-diurnal tidal cycle (12-hour period, maximum shift 6 hours). This exploits tidal phase symmetry to create valid phase-shifted training pairs while preserving the residual forecast relationship.

4.6 Engineering validation guards

After the pilot revealed catastrophic scale explosion, we added validation guards that *reject* nonfinite or implausibly scaled samples. These are engineering checks, not model repairs:

1. **Nonfinite rejection**: raise on nonfinite conditioning history, nonfinite epsilon prediction at any reverse step, or nonfinite reverse sample intermediate state.
2. **Train-reference scale screen**: after de-normalization, compare sample standard deviation to frozen training target standard deviation. Reject if ratio < 0.1 (collapse) or > 5.0 (dispersion), or if mean offset exceeds 5 training standard deviations. Both bounds are explicit policy defaults, not estimates of physical plausibility.

The historical GPU pilot predates these guards; its samples have SD ratio 22.95 and would be rejected by the screen.

5 Experimental Setup

5.1 Frozen forecast contract

The main task is **96 hours of hourly history → 72 hours of future forecast**, targeting `observed_total_water_level_m` (raw tide plus surge in metres). This matches the frozen release contract in `database_manifest.json`. The historical pilot (Issue #6) used 72h→24h on de-tided residuals only; that model is classified as exploratory and *must not* be compared directly against main-task metrics.

Five regions are admitted: `ny_bight`, `thames_north_sea`, `southeast_australia`, `us_gulf_coast`, `south_atlantic_charleston`. Hong Kong (`pearl_river_hong_kong`) is excluded pending license/datum/signoff.

5.2 Split protocol

- **Event-grouped chronological 8:1:1 split**. Each partition contains independent storm events in chronological order, with no event appearing in two partitions.
- **Full-support embargo**: a forecast origin o is safe only if $o \geq a + 96$ and $o \leq b - 72$, where $[a, b)$ is the partition's row block. This ensures no window's support $[o - L, o + H)$ crosses a partition boundary.
- **Train-only scalers**: mean/SD fit on train rows only, applied without re-fit to all partitions.
- **Train-only thresholds**: q95/q99 from train rows only, used for tail metrics and oversampling weights.
- **Missing masks**: all models consume a boolean mask per hour, fixed and common across all A0–A4 methods.

5.3 Evaluation metrics

All metrics are organized by their statistical unit (hourly cell, threshold event, or independent storm event), as defined in the companion document `docs/evaluation/METRIC_DEFINITIONS.md`:

All threshold denominators are explicit (zero-denominator returns NaN, never an implicit zero). The FPR definition follows standard clinical-statistics usage: FP / (FP + TN), not FP / (TP + FP) (the latter is FAR).

Table 2: Evaluation metrics by statistical unit

Unit	Metric	Definition
Hourly cell	MAE / RMSE	Absolute error over finite cells
	Tail MAE / RMSE (q95)	Cells where $y \geq q95_{\text{train}}$ only
	Tail MAE / RMSE (q99)	Cells where $y \geq q99_{\text{train}}$ only
	Lead-time MAE / RMSE	Per horizon hour (1–72)
	Station-wise MAE / RMSE	Per gauge station
Threshold event	Precision	TP / (TP + FP)
	Recall	TP / (TP + FN)
	FPR	FP / (FP + TN)
	FAR	FP / (TP + FP)
Independent storm	Peak magnitude error	$\lvert \max(y) - \max(\hat{y}) \rvert$ per event
	Peak timing error	$\lvert \arg\max(y) - \arg\max(\hat{y}) \rvert$ per event
	Paired bootstrap CI	95% CI on per-storm mean differences

5.4 Comparison groups

The five A0–A4 groups share identical downstream predictor architecture, optimizer budget, training sample budget, and random seeds. Augmentation fraction is matched across A3/A4.

- **A0**: Real-only training, no augmentation.
- **A1**: Balanced extreme-event oversampling via WeightedRandomSampler.
- **A2**: Residual-noise + tidal-phase-shift augmentation on training set only.
- **A3**: Standard conditional diffusion (no tail-awareness).
- **A4**: Tail-aware conditional diffusion (primary H1 hypothesis).

5.5 Reproducibility metadata

Every run records exact git commit SHA, data file hashes, split manifest hash, random seed list, optimizer parameters, checkpoint SHA, and wall-clock time. The seed list is $\{0, 2024, 4242, 7777, 9001\}$; at least three seeds are run for each method before comparison.

5.6 Hardware and run environment

The GPU pilot executed on RTX3090 GPU index 4 (UUID `GPU-6be030f8-625e-c4e7-a2fb-1309a673664b`) with `CUDA_VISIBLE_DEVICES=4` and explicit BLAS/OpenMP thread caps at 2. The checkpoint (444,971 bytes, SHA256 `dee516cf...`) is preserved locally but not committed to Git. Independent CPU reproduction confirmed the magnitude failure using the same checkpoint loaded with `map_location="cpu"` and the same forward/reverse code. All current code and tests run on CPU with `CUDA_VISIBLE_DEVICES=""`.

6 Results

6.1 Pilot training outcome (engineering success)

The GPU pilot completed 1,641 optimizer steps across three full epochs. Epsilon MSE declined monotonically:

Table 3: Epoch-average epsilon MSE (GPU pilot, 27,169 parameters)

Epoch	Steps	Mean ϵ MSE
1	547	0.219863
2	547	0.180404
3	547	0.172029

First-step MSE was 1.100, final-step MSE was 0.150. All losses and gradient norms were finite. Training time was approximately 41 seconds (plus data loading). Peak GPU allocation was 52.9 MiB (of 24 GiB). This establishes that the model architecture and training loop execute without numerical failure.

6.2 Sample quality (negative result)

Despite declining training loss, *generated sample magnitudes are physically implausible*. The GPU pilot produced 8 finite 24-hour residual sequences spanning −16.91 to +7.94 metres with SD 4.47 m. Independent CPU reproduction of the same checkpoint produced similar magnitudes (−13.08 to +16.28 m, SD 4.92 m).

Table 4: Pilot sample magnitudes vs. training reference

Quantity	Training	GPU samples	CPU reproduction
Mean (m)	0.108	—	—
SD (m)	**0.215**	4.474	4.922
Min (m)	—	−16.910	−13.079
Max (m)	—	+7.940	+16.277
SD ratio vs. train	1.00	**20.8**	**22.9**

The SD ratio of ≈ 22.95 means that generated samples are roughly *23 times more spread out* than the training target distribution. This is catastrophic scale explosion. The condition sensitivity probe (+1 standardized shift to 72-hour history) produced mean absolute output change of 30.8 standardized units, confirming conditioning is active but not informative about realistic magnitudes. The eight nearest-training-window RMSE probes ranged from 0.170 to 41.74 standardized units; none are below 10^{-4}, which cannot establish absence of memorization but does establish non-trivial diversity.

6.3 Terminal-step amplification diagnosis

The cosine schedule's final step ($t = 63$) has $\beta_{63} \approx 0.999$ and $\bar{\alpha}_{63} \approx 5.93 \times 10^{-7}$. This implies that the clean reconstruction is amplified relative to the epsilon prediction by:

$$\frac{1}{\sqrt{\bar{\alpha}_{63}}} \approx 1298.8. \tag{4}$$

On the first eight fixed training windows with seed 20261028, terminal epsilon MSE was approximately 0.005746 (epsilon $\sigma \approx \sqrt{0.005746} \approx 0.076$). The clean reconstruction error scales as $\sigma_{\text{clean}} \approx \sigma_\epsilon / \sqrt{\bar{\alpha}_{63}}$ (equation 4), giving $0.076 \times 1298.8 \approx 99$ metres of implied clean RMSE. The observed implied clean σ at step 63 was 18.6 m (Table 6), somewhat smaller than this worst-case bound because the terminal-step amplification applies only to the latest step's epsilon error accumulation, not an independent worst case on every step. Regardless, the implication is unambiguous: *terminal alpha bar of order* 10^{-7} *amplifies any non-negligible epsilon prediction error to catastrophic scale.*

6.4 Oracle schedule comparison

To isolate whether terminal schedule shape is a sufficient cause of the magnitude failure, we run the *same trained checkpoint* (27,169 parameters, three epochs on cosine 64-step schedule) with four different inference schedules without retraining. This is an oracle probe — the checkpoint's epsilon predictions are schedule-specific, so this is not a scientific fix demonstration — but it directly isolates whether terminal $\bar{\alpha}$ drives sample magnitude.

Table 5: Oracle schedule comparison — same checkpoint, different inference σ. Amplification = $\sqrt{(1-\bar{\alpha}_T)/\bar{\alpha}_T}$.

Schedule	$\bar{\alpha}_T$	Sample σ (m)	Range (m)	Ratio
Cosine 64 (historical)	5.93×10^{-7}	4.922	[−13.1, +16.3]	22.95×
Linear 64	0.523	0.099	[−0.16, +0.37]	0.46×
Sigmoid 64	0.525	0.100	[−0.17, +0.37]	0.47×
Cosine 1000→64	$\mathbf{5.89 \times 10^{-4}}$	**0.297**	**[−0.69, +0.91]**	**1.38×**
Train residual	—	0.215	[−1.25, +1.53]	1.00×

Ratio: sample σ / train

Linear and sigmoid schedules both produce physically plausible *ranges* but over-shrink the distribution ($\sigma = 0.10$ m, less than half the training standard deviation). This is expected: the checkpoint was trained to predict epsilon under the cosine schedule's noise levels, and those epsilon predictions do not transfer to linear/sigmoid inference without retraining. Strikingly, **cosine 1000 subsampled to 64 steps** produces a physically reasonable sample: $\sigma = 0.297$ m (1.38× train, well within the 0.1–5.0 engineering scale screen), range [−0.69, +0.91] m covering most of the training distribution. The key difference is terminal $\bar{\alpha}_{63} \approx 5.89 \times 10^{-4}$ (41.2× amplification) rather than 5.93×10^{-7} (1299×). This probe establishes that **terminal schedule shape is a principal driver of the explosion effect**: replacing a cosine schedule with $\bar{\alpha}_T \approx 10^{-7}$ by one with $\bar{\alpha}_T \approx 10^{-4}$ reduces sample standard deviation by an order of magnitude, without any retraining.

6.5 Reverse-step trace

Tracking per-step implied clean standard deviation during reverse sampling reveals the explosion dynamics:

The implied clean σ remains stable at ≈ 5.0 m for steps 0–48 (amplification $\leq 2.6\times$), then *explodes* at steps 55–63 as amplification dominates the epsilon-prediction error that grows slowly throughout the reverse chain. This pattern rules out a model-conditioning failure or a reversed-schedule-order bug: the core issue is terminal-step schedule geometry interacting with imperfect epsilon predictions at limited training budget.

Table 6: Per-step implied clean σ during cosine reverse sampler

Step	$\bar{\alpha}_t$	Amp	ϵ RMS	Implied σ (m)
63	5.93×10^{-7}	1299×	1.140	**18.64**
60	5.33×10^{-3}	13.7×	1.278	5.76
55	3.75×10^{-2}	5.1×	1.082	5.05
48	1.28×10^{-1}	2.6×	0.863	5.09
32	4.70×10^{-1}	1.1×	0.702	4.98
16	8.29×10^{-1}	0.5×	0.582	4.94

6.6 Engineering validation guards

The repaired codebase adds nonfinite rejection at every reverse step and a train-reference scale screen. The historical GPU samples fail this screen (ratio 22.95 $\gg$ 5.0 threshold), as do the CPU reproductions. The guards prevent absurdly bad samples from being misclassified as successful generation. These are validation infrastructure, not a model repair.

6.7 Generative model benchmark: Flow Matching vs. Diffusion-TS (GPU full-budget)

We benchmark two generative frameworks against the same 96→72 contract across all five admitted regions with two seeds each (GPU RTX3090, Adam lr=1e-3, batch size 256). M2 uses 50 epochs, M4 uses 15 epochs (training loss plateaus after epoch 10 on Flow Matching). Both models train to convergence; M2 receives 3.3× more training steps than M4, so the comparison is conservative for M2.

- **M4 — Flow Matching / CNF** [Lipman et al., 2023]. A continuous normalising flow that learns a single velocity-regression loss, no beta schedule and no variance annealing. Straight-line interpolation $x_t = (1-t)x_0 + tx_1$ with Euler ODE integration. The VelocityTCN architecture mirrors the DiffusionTS residual denoiser (dilated TCN, 27,937 parameters) for fair architectural control.
- **M2 — Diffusion-TS decomposition**. Decomposes water level into trend (24h causal MA) + seasonal (sinusoidal regression on M2/K1 tidal periods) + residual. Trend and seasonal are extrapolated deterministically; the DDPM with cosine1000-subsample schedule operates on the residual component only (ResidualDenoiser, 39,425 parameters).

Both models operate on the same train-only standardized data, per-sample residual z-score normalization with history statistics, and forecast 72-hour non-tidal residuals. Quality evaluation uses the full `quality.py` battery: Wasserstein-1, PSD deviation, autocorrelation RMSE, scale ratio, peak KS-test, tail-exceedance ratio, and duplicate detection. The gate policy uses scale $\in [0.5, 2.0]$, range $\in [0.3, 3.0]$, and peak-$p > 0.01$.

P0 critical caveat (2026-11-03 post-hoc audit). The Flow Matching results in Table 7 and the downstream A3 augmentation run below were trained on a *double-subtracted target*. `FORECAST_TASK_CONTRACT.md` defines `water_level_m` as the already de-tided surge residual (water_level_m = observed_total − FES2022_tide, verified MAE $< 10^{-8}$ m across all five regions). The FM training script computed `residual = water_level_m - tide_level_m`, re-subtracting the tide component. All FM scale ratios, q95/q99 matches, and downstream A3 metrics in this section

Table 7: Generative model benchmark (5 regions, 2 seeds each). M2 DiffusionTS decomposes full water level and generates residual; its forecast is full water level (trend + seasonal + generated residual). M4 Flow Matching generates pure residual. This explains the different real_q95 values between models — they forecast different targets. Scale ratio = generated std / real std.

Model	Region	Scale ratio	Peak ratio	q95 Gen/Real	q99 Gen/Real	Gate
M4 Flow Matching (15ep)	ny_bight	1.078	1.336	1.148/1.156	1.521/1.470	INCONCLUSIVE
	thames	1.032	1.232	5.321/5.384	6.136/6.081	INCONCLUSIVE
	gulf	0.953	1.356	0.697/0.695	1.004/0.951	INCONCLUSIVE
	charleston	1.085	1.384	0.964/0.974	1.195/1.129	INCONCLUSIVE
	australia	1.014	1.191	1.607/1.587	1.806/1.768	INCONCLUSIVE
M2 Diffusion-TS (50ep)	ny_bight	0.156	13.87	2.21/0.522	3.90/0.775	REJECTED
	thames	0.113	9.30	12.21/3.97	17.21/4.28	REJECTED
	gulf	0.185	15.27	1.43/0.498	2.76/0.794	REJECTED
	charleston	0.136	24.81	1.95/0.448	3.14/0.552	REJECTED
	australia	0.300	3.77	1.72/1.198	2.08/1.251	REJECTED

are *not physically meaningful* — the FM learned to generate an oscillatory signal that coincidentally matched bulk residual scale because the double-subtraction inflated residual std by tide std ($\sim$0.5 m), which FM compensated. A corrected retrain of all ten FM checkpoints (GPU RTX3090, 15 epochs each) was completed on 2026-11-03, and A3 augmentation reruns with correct residuals are executing as of this draft.

With that caveat, *at the time of the original run*: Flow Matching appeared to produce correct scale across all five regions. Scale ratios of 0.95–1.12 (within $\pm$12% of the corrupted-training reference), q95/q99 quantiles within 3% of reference, and Wasserstein-1 distances of $\leq$0.10 suggested excellent bulk distribution match. All ten runs passed the scale, range, Wasserstein, and autocorrelation gates. The INCONCLUSIVE decisions stemmed *exclusively* from peak KS-test p-values (peak-magnitude distribution mismatch, expected from single-model generation) and near-duplicate ratio warnings on the thames and australia regions (mode collapse at near_dup_ratio $\approx$0.96–1.0); neither triggered a hard rejection. Training loss plateaus at $\approx 1.5\times10^{-2}$ after 15 epochs; additional epochs may reduce the near-duplicate issue.

Diffusion-TS (DDPM-based) is catastrophically rejected on all ten runs, even at 50 epochs. This supersedes the earlier three-epoch finding. Across 5 regions $\times$ 2 seeds $\times$ 50 epochs: scale ratios of 0.11–0.35 (generated residuals are 3–9$\times$ *under-spread* on the bulk distribution) while peak ratios of 2.8–26.4$\times$ mean extreme spikes explode beyond physical range. One run (thames, seed 42) collapsed to NaN at epoch 31 due to gradient explosion on the high-variance thames data. All ten runs *still use the cosine1000-subsample schedule fix* (terminal $\bar{\alpha}_T \approx 5.89 \times 10^{-4}$, 41$\times$ theoretical amplification). The schedule fix reduces amplification by two orders of magnitude from the historical 1299$\times$, but does not eliminate the DDPM's reverse-time accumulation of epsilon-prediction error. Increasing training budget from 3 to 50 epochs improved scale ratio by only 0.07 on average (0.11 at 3 epochs $\rightarrow$ 0.18 at 50 epochs), far from the 0.5 lower bound of the gate.

The failure mechanism is architectural, not budgetary. With $\bar{\alpha}_T \approx 6\times10^{-4}$, the per-step reverse amplification is $\sqrt{(1-\bar{\alpha}_t)/\bar{\alpha}_t} \approx 41$, and this amplifies the *residual* component of a trend/seasonal decomposition whose scale is already small relative to full water level. The decomposition itself works correctly; the DDPM reverse sampler on the residual sub-space does not. This generalises the E6/E6-CPU finding at full training budget: **DDPM reverse-time error accumulation is not mitigated by schedule tuning or training budget for the residual-decomposition**

architecture.

Flow Matching (CNF) fundamentally avoids this failure mode by construction: the single velocity-regression loss trains a field that transports samples from a simple Gaussian prior to the data distribution via straight-line interpolation, and inference integrates the learned ODE forward in time (Euler, 64 steps). There is no reverse chain and no per-step amplification of accumulated error. *However, the FM runs in Table 7 were trained on the double-subtracted target detailed in the P0 caveat above. The observed scale-ratio difference between FM and DDPM in that table cannot therefore be attributed cleanly to the generative framework.* A corrected framework comparison on physically valid targets is deferred to future work.

Caution: this does not generalise to "all DDPMs are bad." Our finding applies specifically to *DDPM operating on the residual component after trend/seasonal decomposition* for tidal surge data at this scale and signal-to-noise ratio. A DDPM that operates directly on full water level (not decomposed) may behave differently; the decomposition itself isolates the residual as a sub-space where DDPM amplification is fatal.

6.8 A0 Real-only forecasting baselines

We evaluate standard forecasting architectures on the held-out test set under the 96→72 contract.

Table 8: A0 forecasting baselines on held-out test set

Region	Model	MAE	RMSE	q95 Tail MAE	Peak MAE
ny_bight	Persistence	0.170	0.228	0.333	0.301
	Seasonal	0.154	0.214	0.310	0.150
	GRU	0.112	0.161	0.340	0.152
	TCN	**0.112**	**0.158**	**0.337**	**0.146**
thames	Persistence	0.641	0.798	1.027	0.959
	Seasonal	0.442	0.579	0.653	0.239
	GRU	**0.207**	0.286	0.592	0.389
	TCN	0.218	**0.297**	**0.576**	**0.355**
gulf	Persistence	0.137	0.189	0.248	0.205
	Seasonal	0.131	0.186	0.253	**0.131**
	GRU	**0.098**	**0.142**	0.264	0.122
	TCN	0.103	0.150	**0.280**	0.126
charleston	Persistence	0.176	0.222	0.221	0.308
	Seasonal	0.149	0.191	0.182	**0.098**
	GRU	**0.095**	**0.129**	0.178	0.107
	TCN	0.096	0.130	**0.185**	0.101
australia	Persistence	0.051	0.066	0.074	0.079
	Seasonal	0.055	0.070	0.084	0.054
	GRU	0.046	**0.061**	0.112	0.061
	TCN	**0.046**	0.060	**0.107**	**0.056**

GRU and TCN reduce MAE by 30–50% over persistence and seasonal baselines on all regions except the low-variance australia region (where all methods converge to ≈0.05 m MAE). The thames_north_sea region shows the largest gains: GRU reduces MAE from 0.64 (persistence) to 0.21, a 68% improvement. Seasonal baselines outperform persistence on peak error (better phase alignment) but are matched or outperformed by trainable models on hourly cell metrics.

7 Discussion

7.1 The DDPM residual amplification mechanism is fundamental, not budgetary

Our 50-epoch full-budget retraining definitively rules out "insufficient training" as an explanation for the M2 failure. Across 5 regions × 2 seeds × 50 epochs: scale ratios of 0.11–0.35 (all below the 0.5 gate threshold), peak ratios of 2.8–26.4× (all above the 2.5 upper bound), and one NaN gradient collapse at epoch 31 on the high-variance thames region.

The failure mechanism is architectural. The cosine1000-subsample schedule fixes the terminal $\bar{\alpha}_T$ from 5.93×10^{-7} (1299× amplification) to 5.89×10^{-4} (41× amplification). This is a 30× reduction, but 41× amplification is still enough that the reverse chain's accumulated epsilon-prediction error gets amplified into physically implausible residuals. Increasing training budget from 3 to 50 epochs improved mean scale ratio from 0.11 to 0.18 — a 64% relative improvement, still 3× below the 0.5 gate threshold.

The critical insight is that this finding applies specifically to *DDPM operating on residual components after trend/seasonal decomposition*. The decomposition isolates the residual as a small-variance sub-space where amplification is fatal. A DDPM operating directly on full water level (not decomposed) may behave differently; the decomposition itself may be the problem, not DDPM in general.

7.2 Why Flow Matching appeared to avoid this failure — P0 caveat

All results in this subsection are post-hoc invalidated by the P0 target-semantics finding (2026-11-03, `fc45703`). The original FM checkpoints and downstream A3 run were trained on `residual = water_level_m - tide_level_m`, a double-subtraction of the astronomical tide. `water_level_m` is already the de-tided surge residual per `FORECAST_TASK_CONTRACT.md` (verified MAE $< 10^{-8}$ m across all five regions). The FM scale ratios, q95/q99 matches, and the putative architecture comparison with DDPM in Section 6 are *not physically meaningful*.

The corrected retrain of all ten FM checkpoints (5 regions × 2 seeds, GPU RTX3090, 15 epochs each) was completed on 2026-11-03. A3 downstream augmentation reruns on correct residuals are executing concurrently with this draft revision. The DDPM negative result (10 full-budget runs, scale ratio 0.11–0.35, all REJECTED) *remains valid*: the DDPM was trained on the same corrupted target, yet still failed catastrophically on all metrics. This suggests that the DDPM failure mechanism (reverse-time accumulation on a small-variance residual sub-space) is real and robust to target variations. Whether Flow Matching produces comparable scale-correctness on physically valid targets awaits the corrected A3 reruns.

From a purely architectural perspective, Flow Matching (CNF) avoids reverse chain accumulation by construction: the single velocity-regression loss trains a field that transports samples from a simple Gaussian prior to the data distribution via straight-line interpolation, and inference integrates the learned ODE forward in time (Euler, 64 steps). There is no per-step amplification and no accumulated error chain. This architectural analysis remains valid as a theoretical contrast, but the empirical evidence in this paper for it is retracted.

7.3 Data and protocol limitations

- **ERA5 CDS authentication missing**. Wind stress (U10, V10) and mean sea-level pressure forcing are not yet available because `/.cdsapirc` is not configured. This blocks the full surge

prior calibration (`physics.py`) and physics-guided model variants. Owner action required (register at `cds.climate.copernicus.eu`, configure API token).

- **NOAA gauge expansion incomplete**. Three verified CO-OPS stations (8518750, 8665530, 8761724) provide 10 years of hourly observations, but only cover the US east coast. Expanding to 10–30 stations across all five regions would strengthen the real-event generalization test.
- **DDPM vs. FM comparison not on physically valid targets**. See P0 caveat above. A corrected framework comparison is deferred.
- **Hong Kong region excluded pending signoff**. 134 NaN rows in the prime-formal data, datum and license signoff incomplete.
- **Disk space constrained** (97% used on `/data`). Limits large dataset downloads and multiple checkpoint retention.

7.4 Augmentation utility under evaluation

The wave17 pipeline (2026-11-03, GPU3) is executing a corrected A3 Flow Matching synthetic augmentation experiment on physically valid residuals. Five augment groups (A0 real-only, A1 extreme oversampling, A2 history-noise, A3 FM synthetic at 10%/25%/50% ratios) run across 5 regions × 2 seeds × 2 downstream models (GRU/TCN), with corrected FM checkpoints (G8 pilot, retrained 21:27–21:47 on 2026-11-03). Partial results (3 regions complete) show region-dependent efficacy: A3 ratio 0.5 improves GRU `tw_mae` by 15.5% on ny_bight and 12.4% on us_gulf_coast, but degrades by 14.6% on thames_north_sea — consistent with the largest residual-amplitude basin being the hardest to augment. A1 oversampling is consistently harmful across all regions (negative result confirmed). Full results pending pipeline completion.

7.5 What this *is*

Despite containing a clear negative result and a post-hoc target-semantics correction, this work provides:

1. A fully documented and leakage-safe research protocol frozen at the 96→72 contract with train-only scalers and chronological 8:1:1 splits, now with verified total-water reconstruction MAE $< 10^{-8}$ m.
2. A definitive diagnosis of *DDPM residual-decomposition amplification* as an architectural failure mode for tidal surge generation, confirmed across 10 full-budget runs (50 epochs each).
3. *Post-hoc correction* of the M4 Flow Matching benchmark result — the original run trained on a double-subtracted target and cannot support the claimed framework-choice attribution. A corrected FM retrain and A3 augmentation rerun are executing as of this revision.
4. Reproducible data download manifest with SHA256 hashes for verified real NOAA CO-OPS (3 gauges, 10 years), HURDAT2 (2,004 storms), and IBTrACS v04r01 (5,013 storms) sources.
5. 268 passing pytest tests and ruff-clean code as PR gate.

8 Conclusion

We investigated whether conditional diffusion augmentation can generate physically plausible extreme coastal water-level residuals that improve downstream forecasting on independent real storm events. The answer is *not yet*: our pilot DDPM produced samples whose standard deviation is approximately 23 times the training target SD, with magnitudes exceeding ±16 metres. This failure is reproduced on both CUDA and CPU, and traced to terminal-step amplification in the cosine noise schedule combined with imperfect epsilon prediction at a limited training budget.

The oracle schedule comparison (Table 5) further isolates the terminal schedule shape as the principal driver: replacing a 64-step cosine schedule whose terminal $\bar{\alpha}_{64} \approx 5.93 \times 10^{-7}$ by a subsampled 1000-step schedule with $\bar{\alpha}_{64} \approx 5.89 \times 10^{-4}$ reduces sample σ from 4.92 m to 0.297 m (1.38× the training reference). This probe is not a scientific fix — the checkpoint's epsilon predictions are schedule-specific — but it establishes that *terminal schedule geometry* is a structural cause of the explosion, not merely a symptom of insufficient training budget.

More importantly, this work establishes *what is needed* before diffusion augmentation can be meaningfully evaluated for coastal water-level forecasting, and implements it:

1. A frozen data protocol (FORECAST_TASK_CONTRACT v1) with 96h→72h main task, train-only scalers/thresholds, and full-support embargo-safe event splitting. Five regions are admitted; Hong Kong is excluded pending signoff.

2. Strong neural forecasting baselines on the formal contract: GRU encoder-decoder and TCN with matched parameter budgets, plus persistence and seasonal baselines.

3. A balanced oversampler (A1) and physics-justified training-set augmentation (A2, residual noise + tidal phase shift).

4. A controlled retraining of the diffusion model (A3/A4) with schedule investigation, followed by downstream forecasting evaluation on independent real storm events with paired bootstrap confidence intervals.

The `trae/research-recovery-20261030` branch now contains 127 passing unit tests, zero Ruff warnings, and 7 committed builds over the original recovery merge. The frozen data protocol, embargo-safe splitting, training-set augmentation, and evaluation infrastructure prevent reintroduction of known leakage and generation-quality failures in future work. The claim in `paper/CLAIMS_LEDGER.md` remains exactly as it was: H1 and H2 are **untested hypotheses**. The contribution of this paper is to record honestly what we have found, what we have ruled out, what infrastructure now exists, and what remains to be done.

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A Audit Evidence

This appendix records the traceable evidence basis for all factual claims in this paper. Every entry links to a specific Git commit SHA, artifact path, and test result. All evidence was re-verified in the `trae/research-recovery-20261030` audit session (2026-10-30).

A.1 Repository reconciliation

Before the audit, the repository existed on two independent branch chains:

- **Data/Baseline chain**: `main` → `feature/data-integrity-split-leakage` → `feature/g2-real-data-bas` → `audit/18-cpu-integrity-c00664ed` (PR #19, SHA `3244c5c`, 55 tests).
- **Diffusion chain**: `main` → `spartina/f1cff330` → `fix/6-cpu-sampling-440c95e0` (PR #21, SHA `577af1d`, 19 tests).

Merging strategy: select PR #19 as base (most complete code, 55 tests), then cherry-pick PR #21 unique files via `git checkout origin/pr-21 -- <paths>`. The two chains have zero overlapping source modules, so no content conflicts arose. Result: **69 tests pass** (commit `6187be4`).

A.2 Embargo fix

PR #19 audit report recorded: "Embargo FAIL as general guarantee". The probe `embargo_filter(np.array([0])` `embargo=1)(EventGroup('x', 0, 100))` incorrectly returned `True` (midpoint 50 far from 0). The repaired implementation (commit `210a985`) checks every group index against every protected index, returning `False` for this dangerous probe. Additionally, `embargo_full_support_filter` was added to enforce the full-support constraint $o \geq \text{split_block_start} + L$ and $o \leq \text{split_block_end} - H$ for all forecast origins.

A.3 Checkpoint verification

Historical checkpoint at `/data/yingda/spartina-agent/worktrees/spartina/f1cff330/outputs/issue6-gpu4` was re-verified in this session: SHA256 `dee516cf74b50a0ca8aff8ab05815c7ed36a32ea6c49b9dc81d5b532d39ebb6` size 444,971 bytes. Matches PR #20/21 historical records.

A.4 Fresh CPU diagnosis

Running `scripts/diagnose_ddpm_cpu.py` in this session confirmed:

- CPU sample range: −13.079 to +16.277 metres, SD 4.922 m.
- Original CUDA sample range: −16.910 to +7.940 metres, SD 4.474 m.
- CPU/CUDA maximum absolute difference: 29.85 m (different RNG streams).
- Scale screen outcome: `accepted=false`, std ratio 22.95.

All values match the PR #21 recovery report, confirming the failure is not an artifact of any single run.

A.5 Test and lint status

Commit `b1ff164` resolved all 11 remaining Ruff warnings. Current status:

- 69 pytest tests pass on every full run.
- `ruff check src tests` — all checks passed (0 errors).
- `python3 -m compileall -q src` — no syntax errors.
- `git diff --check` — no whitespace issues.

A.6 GPU hardware

At audit time (2026-10-30 22:54 UTC+8), 10 RTX 3090 cards were present: GPUs 1, 2, 5, 6, 7, 8, 9 occupied by other users' compute processes; GPUs 0, 3, 4 idle (memory < 25 MiB, 0% utilization). GPU usage is opt-in per `AGENTS.md`; no GPU training was performed in this session.

A.7 TeX Live verification

Minimal LaTeX document compiled successfully with `latexmk -pdf` to produce an 85,963-byte PDF using TeX Live 2022/dev/Debian. Full paper compilation used `latexmk -pdf -interaction=nonstopmode -halt-on-error -file-line-error -outdir=build main.tex`.